

# **homework\_2\_solutions**

## **Microeconomic Analysis**

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**Homework 2**

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## Question 1

Consider the following variants to the data set we used in our first representation theorem (i.e. that a choice correspondence on a finite  $X$  has a utility representation if and only if it satisfies axioms  $\alpha$  and  $\beta$ ). Explain whether or not the statement remains true with these new data sets, and why:

1. There exists some outside option  $o$  which is available for the DM to choose in every choice set.
2. The data set only consists of choice sets with two or three elements.
3. The data set consists only of choice sets of size greater than 2.
4.  $X$  is uncountable, but we only observe choices from all finite subsets of  $X$ .

## Question 1 Solutions

Consider the following variants to the data set we used in our first representation theorem (i.e. that a choice correspondence on a finite  $X$  has a utility representation if and only if it satisfies axioms  $\alpha$  and  $\beta$ ). Explain whether or not the statement remains true with these new data sets, and why:

### Axiom $\alpha$ (Superset to Subset, contraction consistency)

If  $x \in B \subseteq A$  and  $x \in C(A)$ , then  $x \in C(B)$

### Axiom $\beta$ (Subset to superset, expansion consistency)

If  $x, y \in C(B)$ ,  $B \subseteq A$  and  $y \in C(A)$  then  $x \in C(A)$

Combined these must be equivalent to:

Definition of choice correspondence as a utility representation:

$$C(A) = \arg \max_{x \in A} u(x) \quad \text{for every } A \in \mathcal{D}$$

$$C(A) = \{x \in A \mid u(x) \geq u(y) \text{ for every } y \in A\}$$

1. There exists some outside option  $o$  which is available for the DM to choose in every choice set.

The equivalence still holds.

### Necessity ( $\Leftarrow$ ): If a choice correspondence has a utility representation, then $\alpha$ and $\beta$ are satisfied

- Assume there exists a utility function  $u : X \rightarrow \mathbb{R}$  such that  $C(A) = \arg \max_{x \in A} u(x)$  for all  $A$ , which is equivalent to  $C(A) = \{x \in A \mid u(x) \geq u(y) \text{ for every } y \in A\}$ 
  - **Proof of Axiom  $\alpha$ :** Let  $x \in B \subseteq A$  and  $x \in C(A)$ . By the definition of the utility representation,  $u(x) \geq u(y)$  for all  $y \in A$ .

Because  $B$  is a subset of  $A$ , we know that  $u(x) \geq u(y)$  for all  $y \in B$ . Therefore,  $x$  also maximizes  $u$  on  $B$ , implying  $x \in C(B)$  and satisfying  $\alpha$ .

- **Proof of Axiom  $\beta$ :** Let  $x, y \in C(B)$ ,  $B \subseteq A$ , and  $y \in C(A)$ .

Because  $x, y \in C(B)$ , they must both maximize utility in  $B$ , so  $u(x) = u(y) \geq u(z)$  for all  $z \in B$ . Because  $y \in C(A)$ ,  $u(y) \geq u(w)$  for all  $w \in A$ . Since  $u(x) = u(y)$ ,  $u(x) = u(y) \geq u(w)$  for all  $w \in A$ , thus  $u(x) \geq u(w)$  and therefore we have that  $x \in C(A)$ , satisfying  $\beta$ .

- Note that the above does not depend on the existence of any outside option  $o$ . Since the elements and sets are defined arbitrarily, the introduction of an arbitrary element into the potential choices does not affect the necessity proof.

**Sufficiency ( $\Rightarrow$ ):** If  $\alpha$  and  $\beta$  are satisfied, then the choice correspondence has a utility representation

- We will try to partition  $X$  via pairwise choices against  $o$ , such that all elements are either dominated by, indifferent to, or dominating  $o$  and thereby recovering the typical pairwise comparisons of choices and preventing cyclicity across and within the partitions.
- First, assume  $C(\cdot)$  satisfies Axioms  $\alpha$  and  $\beta$  on  $\mathcal{D} = \{A \subseteq X \mid o \in A\}$ .
- Then, for any  $x \in X$ , the pairwise set  $B_x = \{x, o\}$  is contained in  $\mathcal{D}$ . We can partition  $X$  into three mutually disjoint subsets based on the choice from  $B_x$ :

**1. Strictly dominating elements:**

$$X^+ = \{x \in X \setminus \{o\} \mid C(\{x, o\}) = \{x\}\}$$

**2. Indifferent elements:**  $X^= = \{x \in X \setminus \{o\} \mid C(\{x, o\}) = \{x, o\}\}$

**3. Strictly dominated elements:**

$$X^- = \{x \in X \setminus \{o\} \mid C(\{x, o\}) = \{o\}\}$$

- Next, we establish hierarchical dominance across the sets to prove that cyclical choices between the disjoint subsets are impossible.

- Clearly, all elements in  $X^-$  are never chosen, since they are strictly dominated by  $o$  and  $o$  is always available as a choice.
- Likewise, all elements in  $X^+$  strictly dominate elements in  $X^-$ ,  $X^=$  and  $\{o\}$  by construction of the set, so elements from  $X^+$  are always chosen over elements from the other subsets.
- Finally,  $\{o\}$  forms an equivalence class with  $X^=$ , such that all elements within that subset are indifferent with each other and with  $o$ .
- We must verify that preferences are complete within the strictly dominant set  $X^+$ . If we define the preference relation  $\succeq$  on  $X^+$  such that  $x \succeq y \iff x \in C(\{x, y, o\})$ , then since  $C(\{x, y, o\})$  cannot contain  $o$  since we are in  $X^+$ , it must contain  $x$  or  $y$ , establishing completeness on  $X^+$ .
- Next, to establish transitivity within  $X^+$ , assume  $x \succeq y$  and  $y \succeq z$ . Consider  $S = \{x, y, z, o\}$ . Again since we are in  $X^+$ ,  $o \notin C(S)$ . Thus,  $C(S) \subseteq \{x, y, z\}$ .
  1. If  $z \in C(S)$ , Axiom  $\alpha \implies z \in C(\{y, z, o\})$ . Because  $y \succeq z$ ,  $y \in C(\{y, z, o\})$ . Axiom  $\beta \implies y \in C(S)$ .
  2. If  $y \in C(S)$ , Axiom  $\alpha \implies y \in C(\{x, y, o\})$ . Because  $x \succeq y$ ,  $x \in C(\{x, y, o\})$ . Axiom  $\beta \implies x \in C(S)$ .
- Since  $C(S)$  must be non-empty, from above we know that  $x \in C(S)$  in all cases. By Axiom  $\alpha$ ,  $x \in C(S) \implies x \in C(\{x, z, o\})$ , establishing  $x \succeq z$ .
- Therefore we have that  $\succeq$  is complete and transitive on  $X^+$  with acyclicity for the strictly dominant subset and therefore must have a utility function  $u^*(x)$  that represents it, such that  $x \succsim y \implies u^*(x) \geq u^*(y) \forall x, y \in X^+$ .
- Extending to the rest of  $X$ , we know that any element  $x^* \in X^+$  will strictly dominate any element of  $X^=$  or  $X^-$ , such that we can form a strict preference relation  $x^- \succ x^= \succ x^*$  for any  $x^- \in X^-$  and any

$x^- \in X^-$ , which includes  $o \succ x^*$ , ensuring that all elements of  $X$  are comparable (complete) and preferences are transitive (no cycles).

- Thus since  $\succsim$  is a complete and transitive relation on  $X$ , we can construct a utility function  $u : X \rightarrow \mathbb{R}$  that represents it:

$$u(x) = \begin{cases} u^*(x) & \text{if } x \in X^+ \\ 0 & \text{if } x \in X^- \cup \{o\} \\ -1 & \text{if } x \in X^- \end{cases}$$

- Therefore we have shown that even in the case of an outside option  $o$ , as long as the axioms  $\alpha$  and  $\beta$  hold, the choice correspondence has a representative utility function consistent with the preference relation.

2. The data set only consists of choice sets with two or three elements.

The equivalence still holds.

**Necessity** ( $\Leftarrow$ ): If a choice correspondence has a utility representation, then  $\alpha$  and  $\beta$  are satisfied

- Assume there exists a utility function  $u : X \rightarrow \mathbb{R}$  such that  $C(A) = \arg \max_{x \in A} u(x)$  for all  $A$ , which is equivalent to  $C(A) = \{x \in A \mid u(x) \geq u(y) \text{ for every } y \in A\}$ 
  - **Proof of Axiom  $\alpha$ :** Let  $x \in B \subseteq A$  and  $x \in C(A)$ . By the definition of the utility representation,  $u(x) \geq u(y)$  for all  $y \in A$ . Because  $B$  is a subset of  $A$ , we know that  $u(x) \geq u(y)$  for all  $y \in B$ . Therefore,  $x$  also maximizes  $u$  on  $B$ , implying  $x \in C(B)$  and satisfying  $\alpha$ .
  - **Proof of Axiom  $\beta$ :** Let  $x, y \in C(B)$ ,  $B \subseteq A$ , and  $y \in C(A)$ . Because  $x, y \in C(B)$ , they must both maximize utility in  $B$ , so  $u(x) = u(y) \geq u(z)$  for all  $z \in B$ . Because  $y \in C(A)$ ,  $u(y) \geq u(w)$  for all  $w \in A$ . Since  $u(x) = u(y)$ ,  $u(x) = u(y) \geq u(w)$  for all  $w \in A$ , thus  $u(x) \geq u(w)$  and therefore we have that  $x \in C(A)$ , satisfying  $\beta$ .
- Note that the above does not depend on a restriction in the number of elements in the data. Since the elements and sets are defined arbitrarily, limiting the number of potential choices does not affect the necessity proof.

**Sufficiency** ( $\Rightarrow$ ): If  $\alpha$  and  $\beta$  are satisfied, then the choice correspondence has a utility representation

- Assume  $C$  satisfies Axioms  $\alpha$  and  $\beta$ . Define a preference relation and choice correspondence:  $x \succsim y \iff x \in C(\{x, y\})$ .
- **Completeness:** For all  $x, y \in X$ , since choice sets must be non-empty, either  $x \in C(\{x, y\})$  or  $y \in C(\{x, y\})$ .

- *Transitivity*: Consider the sets of two elements. Without loss of generality, we can define  $\{x, y\}$  as satisfying  $x \succsim y$  and  $\{y, z\}$  as satisfying  $y \succsim z$ . If there exists a superset  $A = \{x, y, z\}$ , then we examine the following cases; we know that at least one element must be chosen from the superset:
  - If  $z \in C(A)$ , Axiom  $\alpha$  implies  $z \in C(\{y, z\})$ . Because  $y \succsim z$ , we know that  $y \in C(\{y, z\})$ , and Axiom  $\beta$  implies  $y \in C(A)$ .
  - Either from case 1 or by its own assumption, if  $y \in C(A)$ , since  $x \succsim y$ , Axiom  $\beta$  implies  $x \in C(A)$ .
  - Finally, if  $x \in C(A)$ , then we must have that  $x \succsim z$ , such that the preference relation is transitive.
- Completeness and transitivity hold on  $\succsim$  for all the sets of two elements and the superset of three elements, therefore the preference relation has a utility representation.

3. The data set consists only of choice sets of size greater than 2.

The equivalence still holds.

**Necessity** ( $\Leftarrow$ ): If a choice correspondence has a utility representation, then  $\alpha$  and  $\beta$  are satisfied

- Assume there exists a utility function  $u : X \rightarrow \mathbb{R}$  such that  $C(A) = \arg \max_{x \in A} u(x)$  for all  $A$ , which is equivalent to  $C(A) = \{x \in A \mid u(x) \geq u(y) \text{ for every } y \in A\}$ 
  - **Proof of Axiom  $\alpha$ :** Let  $x \in B \subseteq A$  and  $x \in C(A)$ . By the definition of the utility representation,  $u(x) \geq u(y)$  for all  $y \in A$ . Because  $B$  is a subset of  $A$ , we know that  $u(x) \geq u(y)$  for all  $y \in B$ . Therefore,  $x$  also maximizes  $u$  on  $B$ , implying  $x \in C(B)$  and satisfying  $\alpha$ .
  - **Proof of Axiom  $\beta$ :** Let  $x, y \in C(B)$ ,  $B \subseteq A$ , and  $y \in C(A)$ . Because  $x, y \in C(B)$ , they must both maximize utility in  $B$ , so  $u(x) = u(y) \geq u(z)$  for all  $z \in B$ . Because  $y \in C(A)$ ,  $u(y) \geq u(w)$  for all  $w \in A$ . Since  $u(x) = u(y)$ ,  $u(x) = u(y) \geq u(w)$  for all  $w \in A$ , thus  $u(x) \geq u(w)$  and therefore we have that  $x \in C(A)$ , satisfying  $\beta$ .
- Note that the above does not depend on a restriction in the number of elements in the data. Since the elements and sets are defined arbitrarily, limiting the data set to choice sets of size greater than 2 does not affect the necessity proof.

**Sufficiency** ( $\Rightarrow$ ): If  $\alpha$  and  $\beta$  are satisfied, then the choice correspondence has a utility representation

- Assume the choice  $C$  satisfies Axioms  $\alpha$  and  $\beta$  for all sets  $A$  such that  $|A| > 2$ . Define the finite set  $X^* = \{x \in X \mid x \in C(A)\}$  and preference relation  $\succeq$  such that  $x \succeq y$  for  $x, y \in A$  and  $x \in C(A)$ . We aim to show

that the preference relation is complete and transitive on  $X^*$  and therefore has a utility representation  $u$ .

- **Completeness:** Consider the superset  $S = A \cup \{y\}$  for some  $A \subseteq X$  and  $y \in X$ . Because  $|A| \geq 3$ , it follows that  $|S| \geq 3$  and is thus also one of the potential domain sets. Since choice sets must yield a choice,  $C(S)$  is non-empty. Let  $w \in C(S)$ . We evaluate the three possible cases for  $w$ :
  - If  $w = x$ , then  $x \in C(S)$ . Since  $y \in S$ , by definition we have  $x \succeq y$ .
  - If  $w = y$ , then  $y \in C(S)$ . Since  $x \in S$ , we have  $y \succeq x$ .
  - If  $w \neq x$  and  $w \neq y$ , then  $w \in A$ . By Axiom  $\alpha$ ,  $w \in C(S)$  and  $w \in A \subseteq S$  implies  $w \in C(A)$ . We now have  $x, w \in C(A)$ . By Axiom  $\beta$ , since  $w \in C(S)$  and  $A \subseteq S$ , it must be that  $x \in C(S)$ . Because  $x \in C(S)$  and  $y \in S$ , we have  $x \succeq y$ .
  - In all cases, either  $x \succeq y$  or  $y \succeq x$ . Therefore, the relation  $\succeq$  is complete on  $X^*$ .
- **Transitive:** Assume  $x \succeq y$  and  $y \succeq z$ . By definition, there exists  $A$  where  $x \in C(A)$  and  $y \in A$ , and there exists  $B$  where  $y \in C(B)$  and  $z \in B$ .
  - Consider the superset  $S = A \cup B$ . Because  $|A| \geq 3$ ,  $|S| \geq 3$ . Let  $w \in C(S)$ .
  - If  $w \in A$ : By Axiom  $\alpha$ ,  $w \in C(A)$ . Because we know  $x \in C(A)$ , Axiom  $\beta$  implies  $x \in C(S)$ . Since  $z \in B \subseteq S$ , we have  $x \in C(S)$  and  $z \in S$ , establishing  $x \succeq z$ .
  - If  $w \in B$ : By Axiom  $\alpha$ ,  $w \in C(B)$ . Because we know  $y \in C(B)$ , Axiom  $\beta$  implies  $y \in C(S)$ . Since  $y \in C(S)$  and  $y \in A$ , Axiom  $\alpha$  implies  $y \in C(A)$ . Because  $x \in C(A)$ , Axiom  $\beta$  implies  $x \in C(S)$ . Thus we have  $x \succeq z$ .
  - Transitivity holds in all cases.
- Because  $\succeq$  is a complete and transitive binary relation on the finite set  $X^*$ , there exists a utility function  $u : X^* \rightarrow \mathbb{R}$  such that  $x \succeq y \iff u(x) \geq u(y)$ .

4.  $X$  is uncountable, but we only observe choices from all finite subsets of  $X$ .

The statement no longer holds for this restriction since  $\alpha$  and  $\beta$  are not sufficient to prove the existence of a utility representation for uncountable sets, regardless of whether we observe choices from finite subsets.

**Necessity** ( $\Leftarrow$ ): If a choice correspondence has a utility representation, then  $\alpha$  and  $\beta$  are satisfied

- Assume there exists a utility function  $u : X \rightarrow \mathbb{R}$  such that  $C(A) = \arg \max_{a \in A} u(a)$  for all  $A \in \mathcal{D}$ .
- Let  $x \in B \subseteq A$  with  $A, B \in \mathcal{D}$  and  $x \in C(A)$ . By the definition of the representation,  $u(x) \geq u(z)$  for all  $z \in A$ . Because  $B$  is a subset of  $A$ , it follows that  $u(x) \geq u(z)$  for all  $z \in B$ . Therefore,  $x$  maximizes  $u$  on  $B$ , implying  $x \in C(B)$ . Axiom  $\alpha$  holds unconditionally.
- Let  $x, y \in C(B)$ ,  $B \subseteq A$ , and  $y \in C(A)$  with  $A, B \in \mathcal{D}$ . Since  $x, y \in C(B)$ ,  $u(x) = u(y)$ . Since  $y \in C(A)$ ,  $u(y) \geq u(z)$  for all  $z \in A$ . By transitivity of equality and inequalities in  $\mathbb{R}$ ,  $u(x) \geq u(z)$  for all  $z \in A$ , forcing  $x \in C(A)$ . Axiom  $\beta$  holds unconditionally.

**Sufficiency** ( $\Rightarrow$ ): If  $\alpha$  and  $\beta$  are satisfied, then the choice correspondence has a utility representation

- The statement does not hold in this direction.
- Assume a choice correspondence  $C$  satisfies Axioms  $\alpha$  and  $\beta$  for all finite subsets of  $X$ , such that an uncountable set  $X$  contains all base pairs  $\{x, y\}$  and triples  $\{x, y, z\}$ . Define the preference relation  $x \succsim y \iff x \in C(\{x, y\})$ . We would normally aim to show that with Axioms  $\alpha$  and  $\beta$ , the pairs and triples under  $\succsim$  form a complete and transitive preference relation on  $X$ .
- However, a complete and transitive preference relation on an uncountable set  $X$  is insufficient to guarantee a real-valued utility

representation  $u : X \rightarrow \mathbb{R}$ . Uncountably infinite sets like the reals also require preference relations to satisfy order separability to have a utility function, which are not guaranteed by Axioms  $\alpha$  and  $\beta$ .

- Consider the following counterexample on  $X = \mathbb{R}^2$  with the lexicographic preference

$$(x_1, x_2) \succsim (y_1, y_2) \iff x_1 > y_1, \text{ or } (x_1 = y_1 \text{ and } x_2 \geq y_2).$$

- For any finite set  $A \subset \mathbb{R}^2$ , a unique max exists, which we can call  $C(A)$ , which satisfies Axioms  $\alpha$  and  $\beta$  since it imposes an order on every subset.
- But suppose for contradiction that  $C(*)$  is represented by a utility function  $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ . For any real number  $x \in \mathbb{R}$ ,  $(x, 1) \succ (x, 0)$ , which implies  $u(x, 1) > u(x, 0)$ .
- Because the rational numbers  $\mathbb{Q}$  are dense in  $\mathbb{R}$ , for each  $x \in \mathbb{R}$ , there exists a rational number  $q_x \in \mathbb{Q}$  such that  $u(x, 1) > q_x > u(x, 0)$ .
- But if  $x > y$ , we must have that  $(x, 0) \succ (y, 1)$ , implying  $u(x, 0) > u(y, 1)$ . Together, this gives  $q_x > u(x, 0) > u(y, 1) > q_y \implies q_x > q_y$ .
- Thus for every distinct  $x$ , there exists a distinct rational  $q_x$ , which implies a one-to-one mapping from the uncountable set  $\mathbb{R}$  into the countable set  $\mathbb{Q}$ , which is impossible.
- Therefore, we have arrived at a contradiction, which means that our assumption that a utility function  $u : X \rightarrow \mathbb{R}$  representing the choice correspondence exists was false.
- Therefore, since sufficiency fails, the uncountably infinite set  $X$  does not permit a utility function representation even though it satisfies  $\alpha$  and  $\beta$ . Observing finite subsets of  $X$  cannot recover the utility function since it does not resolve the one-to-one mapping issue.

## Question 2

Some warm up questions:

1. Show that the three definitions of continuity we introduced in class are identical.
2. Can a continuous preference relation be represented by a discontinuous utility function?
3. Show that any preference relation on a set  $X$  that can be represented by some utility function  $u : X \rightarrow \mathbb{R}$  can also be represented by a utility function  $v : X \rightarrow [0, 1]$ .
4. Let  $\succsim$  be continuous preferences on a set  $X \subseteq \mathbb{R}^n$  that contains the interval connecting the points  $x$  and  $z$ . Show that if  $x \succsim y \succsim z$  then there is a point  $m$  on the interval connecting  $x$  and  $z$  such that  $y \sim m$ .

## Question 2 Solutions

Some warm up questions:

1. Show that the three definitions of continuity we introduced in class are identical.

The three definitions of continuity are:

1. We say that a preference relation  $\succsim$  on a metric space  $X$  is continuous if, for any  $x, y \in X$  such that  $x \succ y$ , there exists an  $\epsilon > 0$  such that, for any  $x' \in B(x, \epsilon)$  and  $y' \in B(y, \epsilon)$ ,  $x' \succ y'$ .
2. A preference relation  $\succsim$  on a metric space  $X$  is continuous if and only if, for every  $x, y \in X$  and sequence  $\{x_n, y_n\}$  such that  $x_n \rightarrow x$  and  $y_n \rightarrow y$  then  $x_n \succsim y_n \forall n$  implies  $x \succsim y$ .
3. A preference relation  $\succsim$  on a metric space  $X$  is continuous if and only if, for any  $x$  the sets  $\{y | y \succ x\}$  and  $\{y | x \succ y\}$  are open.

First, to show that 1 implies 2:

- Assume Definition 1 holds: for any  $x, y \in X$  such that  $x \succ y$ , there exists an  $\epsilon > 0$  such that for all  $x' \in B(x, \epsilon)$  and  $y' \in B(y, \epsilon)$ ,  $x' \succ y'$ .
- Let  $\{x_n\}$  and  $\{y_n\}$  be sequences in  $X$  such that  $x_n \rightarrow x$  and  $y_n \rightarrow y$ , with the property that  $x_n \succsim y_n$  for all  $n$ . We must show that  $x \succsim y$ .
- Assume for contradiction that  $x \not\succsim y$  and therefore  $y \succ x$ . By Definition 1, there exists an  $\epsilon > 0$  such that for any  $y' \in B(y, \epsilon)$  and  $x' \in B(x, \epsilon)$ ,  $y' \succ x'$ .
- Because  $\{x_n\}$  and  $\{y_n\}$  converge, there exists an integer  $N$  such that for all  $n \geq N$ ,  $y_n \in B(y, \epsilon)$  and  $x_n \in B(x, \epsilon)$ . Therefore, for all  $n \geq N$ , it must be true that  $y_n \succ x_n$ .
- But this contradicts that  $x_n \succsim y_n$  for all  $n$ . Therefore, the assumption  $y \succ x$  must be false, forcing  $x \succsim y$ . Therefore we have satisfied Definition 2.

Then to show that 2 implies 3:

- Assume Definition 2 holds: for every  $x, y \in X$  and sequence  $\{x_n, y_n\}$  such that  $x_n \rightarrow x$  and  $y_n \rightarrow y$  then  $x_n \succsim y_n \forall n$  implies  $x \succsim y$ .
- We must show that for any  $x \in X$ , the sets  $\{y | y \succ x\}$  and  $\{y | x \succ y\}$  are open. These are just the strict upper contour set  $U(x) = \{y \in X | y \succ x\}$  and the strict lower contour set  $L(x) = \{y \in X | x \succ y\}$ . By definition, a set is open if and only if its complement in  $X$  is closed.
- Thus let us consider the complement of the strict upper contour set  $X \setminus U(x) = \{y \in X | y \not\succ x\}$ . This is equivalent to the weak lower contour set  $\{y \in X | x \succsim y\}$ .
  - Let  $\{y_n\}$  be any sequence in  $X \setminus U(x)$  that converges to a limit  $y \in X$ .
  - Define a constant sequence  $x_n = x$  for all  $n$ , which clearly converges to  $x$ .
  - Since  $y_n \in X \setminus U(x)$ , we have  $x_n \succsim y_n$  for all  $n$ . By Definition 2, the limits must satisfy  $x \succsim y$ , such that  $y \in X \setminus U(x)$ .
  - Since the preference relation  $\succsim$  is closed under sequential limits, the weak contour set  $\{y \in X | x \succsim y\}$  is closed. As we said,  $X \setminus U(x) = \{y \in X | y \not\succ x\}$  is equivalent to the weak lower contour set  $\{y \in X | x \succsim y\}$ .
  - So since the weak lower contour set  $\{y \in X | x \succsim y\}$  is closed, thus  $X \setminus U(x) = \{y \in X | y \not\succ x\}$  is closed.
  - Since the complement  $X \setminus U(x) = \{y \in X | y \not\succ x\}$  is closed, we must have that  $U(x)$  is open.
- By a symmetric argument, the complement of the strict lower contour set  $X \setminus L(x) = \{y \in X | y \not\succ x\}$ . Taking a similar sequence  $y_n \succsim x_n$  converging to  $y$ , Definition 2 tells us that  $y \succsim x$ , confirming the complement is closed and  $L(x)$  is open.

- Thus we have that if Definition 2 holds, the strict upper contour sets  $\{y|y \succ x\}$  and  $\{y|x \succ y\}$  are open.

Finally, to show that 3 implies 1:

- Assume Definition 3 holds: for any  $x$  the sets  $\{y|y \succ x\}$  and  $\{y|x \succ y\}$  are open. To avoid notational confusion, we redefine this definition as: for any  $z \in X$ , the strict contour sets  $U(z) = \{w \in X \mid w \succ z\}$  and  $L(z) = \{w \in X \mid z \succ w\}$  are open in  $X$ .
- Let  $x, y \in X$  such that  $x \succ y$ . To prove definition 1, we must find an  $\epsilon > 0$  such that for any  $x' \in B(x, \epsilon)$  and  $y' \in B(y, \epsilon)$ ,  $x' \succ y'$ .
- Consider the following cases:
  - Case A: An intermediate element  $z$  exists
    - Suppose there exists some  $z \in X$  strictly between  $x$  and  $y$ , such that  $x \succ z \succ y$ . Thus,  $x \in U(z)$  and  $y \in L(z)$ .
    - Because both contour sets are open by Definition 3, there exist  $\epsilon_x > 0$  and  $\epsilon_y > 0$  such that  $B(x, \epsilon_x) \subseteq U(z)$  and  $B(y, \epsilon_y) \subseteq L(z)$ .
    - Let  $\epsilon = \min(\epsilon_x, \epsilon_y)$ . For any  $x' \in B(x, \epsilon)$  and  $y' \in B(y, \epsilon)$ , we have  $x' \in U(z) \implies x' \succ z$ , and  $y' \in L(z) \implies z \succ y'$ .
    - By transitivity,  $x' \succ y'$ , so this case satisfies Definition 1.
  - Case B: No intermediate element  $z$  exists
    - Suppose no such  $z$  exists. Then for any  $w \in X$ , either  $w \succsim x$  or  $y \succsim w$  and the space partitions perfectly into the strict contour sets of  $x$  and  $y$ :  $X = U(y) \cup L(x)$ .
    - This implies that the intersection of the contour sets is empty  $U(y) \cap L(x) = \emptyset$ , because if there were any element  $w \in U(y) \cap L(x)$  it would imply  $x \succ w \succ y$ , violating our assumption.
    - Assume for contradiction that no such  $\epsilon > 0$  exists to separate  $x$  and  $y$ .

- Then for every  $n \in \mathbb{N}$ , there exist elements  $x_n \in B(x, 1/n)$  and  $y_n \in B(y, 1/n)$  such that  $y_n \succsim x_n$ . By construction,  $x_n \rightarrow x$  and  $y_n \rightarrow y$ .
- Since  $x \succ y$ , we know  $x \in U(y)$ . Because  $U(y)$  is an open set and  $x_n \rightarrow x$ , for sufficiently large  $n$ ,  $x_n \in U(y) \implies x_n \succ y$ .
- Since  $x \succ y$ , we also know  $y \in L(x)$ . Because  $L(x)$  is an open set and  $y_n \rightarrow y$ , we must have that  $y_n \in L(x) \implies x \succ y_n$ .
- Thus we have  $x \succ y_n$  and  $y_n \succsim x_n$ , which requires  $x \succ x_n$ . But we said earlier that  $x_n \in U(y)$ . Since  $U(y) \cap L(x) = \emptyset$ ,  $x_n$  cannot be in both  $U(y)$  and  $L(x)$ . Thus we have arrived at a contradiction.
- Therefore, our initial assumption must be false, and therefore  $\epsilon > 0$  must exist and Definition 1 holds.

Since we have shown that Definition 1 implies Definition 2, Definition 2 implies Definition 3, and Definition 3 implies Definition 1, we must have that these three definitions of continuity are equivalent, since if any one of them holds the entire cycle must hold.

## 2. Can a continuous preference relation be represented by a discontinuous utility function?

- Let  $\succsim$  be a continuous preference relation on  $X$ . By representation theorem if  $X$  is separable, there exists a continuous utility function  $u : X \rightarrow \mathbb{R}$  such that for all  $x, y \in X$ ,  $x \succsim y \iff u(x) \geq u(y)$ .
- If  $f : \mathbb{R} \rightarrow \mathbb{R}$  is a strictly increasing function, the composite function  $v = f \circ u$  preserves the preference ranking  
 $v(x) \geq v(y) \iff f(u(x)) \geq f(u(y)) \iff u(x) \geq u(y) \iff x \succsim y$
- To show if the utility function can be discontinuous, we have to construct a strictly increasing  $f$  with a jump discontinuity.
  - Assume the image of  $X$  under  $u$  includes  $[0, 1]$ . Define a transformation  $f : \mathbb{R} \rightarrow \mathbb{R}$  as:

$$f(z) = \begin{cases} z & \text{if } z < 0.5 \\ z + 1 & \text{if } z \geq 0.5 \end{cases}$$

- $f$  is strictly monotonic, ensuring  $v(x) = f(u(x))$  is a valid representation of  $\succsim$ . So now we have to show that  $v(x)$  satisfies continuity.
- By definition, a function is continuous if and only if the preimage of every open set in the codomain is an open set in the domain.
- Consider the open interval  $(0.8, 1.8)$  in the codomain  $\mathbb{R}$ . The preimage of this open set under  $v$  is  $v^{-1}((0.8, 1.8)) = \{x \in X \mid 0.8 < v(x) < 1.8\}$ .
- $v(x)$  only falls within the jump discontinuity of  $f$  when  $1.5 \leq v(x) < 1.8$ . Mapping back through  $f$  this corresponds to  $0.5 \leq u(x) < 0.8$ .
- Thus the preimage is  $u^{-1}([0.5, 0.8))$ . Because the preimage is partly closed, due to the discontinuity in  $f$ , it is not an open set. Since the preimage of the open set  $(0.8, 1.8)$  is not open in the domain, the function  $v(x)$  is not continuous.

- However, since we have shown that  $v(x)$  is still strictly increasing, it preserves the preference rankings from  $u(x)$ , such that for  $x, y \in X$ ,  $x \succsim y \iff v(x) \geq v(y)$ .
- Therefore, we have shown that a continuous preference relation can be represented by a discontinuous utility function.

3. Show that any preference relation on a set  $X$  that can be represented by some utility function  $u : X \rightarrow \mathbb{R}$  can also be represented by a utility function  $v : X \rightarrow [0, 1]$ .

Let  $\succsim$  be a preference relation on a set  $X$  that is represented by a utility function  $u : X \rightarrow \mathbb{R}$ . By the definition of a utility representation, for all  $x, y \in X$ :

$$x \succsim y \iff u(x) \geq u(y)$$

To prove that  $\succsim$  can also be represented by a utility function mapping to  $[0, 1]$ , we must construct a monotonically increasing transformation  $f : \mathbb{R} \rightarrow [0, 1]$ . Because ordinal utility representations are invariant under monotonically increasing transformations, the composite function  $v = f \circ u$  will preserve the exact preference ordering.

Define  $f : \mathbb{R} \rightarrow [0, 1]$  as  $f(z) = \frac{1}{1+e^{-z}}$  which has a range of  $(0, 1)$  and thus is contained in  $[0, 1]$ . To show  $f$  is monotonically increasing, we can take the first derivative  $f'(z) = \frac{e^{-z}}{(1+e^{-z})^2}$ . Since  $e^{-z} > 0$  for all  $z \in \mathbb{R}$ ,  $f$  is monotonically increasing.

Thus we have  $v = f \circ u = f(u(x)) = \frac{1}{1+e^{-u(x)}}$  forms a valid utility function mapping  $X \rightarrow [0, 1]$  and we can show that

$$v(x) \geq v(y) \iff \frac{1}{1+e^{-u(x)}} \geq \frac{1}{1+e^{-u(y)}} \iff x \succsim y.$$

4. Let  $\succsim$  be continuous preferences on a set  $X \subseteq \mathbb{R}^n$  that contains the interval connecting the points  $x$  and  $z$ . Show that if  $x \succsim y \succsim z$  then there is a point  $m$  on the interval connecting  $x$  and  $z$  such that  $y \sim m$ .
- Let  $S = \{\alpha x + (1 - \alpha)z \mid \alpha \in [0, 1]\}$ ,  $S \subseteq X$  be the closed line segment connecting  $x$  and  $z$ . We aim to show that  $y \sim m$  using sequential convergence using two sequences  $\{a_k\}$  and  $\{b_k\}$  in  $S$ .
  - Let  $a_0 = x$  and  $b_0 = z$ . Since  $x \succsim y \succsim z$ , we know that  $a_0 \succsim y \succsim b_0$ .
  - First we can find the midpoint of the segment  $m_k = \frac{1}{2}a_k + \frac{1}{2}b_k$
  - Because  $\succsim$  is complete, either  $m_k \succsim y$  or  $y \succsim m_k$ . We can update the sequences to the midpoint of each smaller sequence while preserving the preference relation as follows:
    - If  $m_k \succsim y$ , define  $a_{k+1} = m_k$  and  $b_{k+1} = b_k$ .
    - If  $y \succsim m_k$  (and  $m_k \not\succsim y$ ), define  $a_{k+1} = a_k$  and  $b_{k+1} = m_k$ .
  - For all  $k \in \mathbb{N}$ , the terms satisfy  $a_k \succsim y$  and  $y \succsim b_k$ .
  - We can use the Euclidean norm since we are in the real numbers to find the distance between any  $a_k$  and  $b_k$  is given by
 
$$\|a_k - b_k\| = \frac{1}{2^k} \|x - z\|.$$
  - Since we are halving the distance each time, as  $k \rightarrow \infty$ ,  $\|a_k - b_k\| \rightarrow 0$ . Because  $\{a_k\}$  and  $\{b_k\}$  are bounded sequences within the compact subspace  $S \subset \mathbb{R}^n$ , and their distance converges to zero, they are Cauchy and therefore converge to a shared limit point  $m$ . Because  $S$  is closed,  $m \in S$ .
  - Therefore,  $\lim_{k \rightarrow \infty} a_k = m$  and  $\lim_{k \rightarrow \infty} b_k = m$ .
  - Using Definition 2 (sequential definition) of continuous preferences, we know preference is preserved in the limit.
    - Because  $a_k \succsim y$  for all  $k$ , and  $a_k \rightarrow m$ , sequential continuity guarantees  $m \succsim y$ .
    - Because  $y \succsim b_k$  for all  $k$ , and  $b_k \rightarrow m$ , sequential continuity guarantees  $y \succsim m$ .

- Since  $m \succsim y$  and  $y \succsim m$ , we must have that  $y \sim m$ .

### Question 3

Let  $X$  be a non-empty set and  $\succsim$  be an order separable preference relation.

1. Show that this implies that there exists a countable set  $Y \subset X$  such that, for any  $x \succ z$ , there exists and  $y_1$  and  $y_2$  in  $Y$  such that

$$x \succsim y_1 \succ y_2 \succsim z$$

2. Let  $Y$  be such a set numbered  $\{y_1, y_2, \dots\}$ . Define

$L(x) = \{m \in \mathbb{N} \mid x \succ y_m\}$ . Define  $u : X \rightarrow [0, 1]$  as

$$u(x) = \begin{cases} 0 & \text{if } L(x) \text{ is empty} \\ \sum_{m \in L(x)} \frac{1}{2^m} & \text{otherwise} \end{cases}$$

Show  $u$  represents  $\succsim$ .

## Question 3 Solutions

Let  $X$  be a non-empty set and  $\succsim$  be an order separable preference relation.

1. Show that this implies that there exists a countable set  $Y \subset X$  such that, for any  $x \succ z$ , there exists and  $y_1$  and  $y_2$  in  $Y$  such that

$$x \succsim y_1 \succ y_2 \succsim z$$

- Let  $\succsim$  be a complete, transitive, and order separable preference relation on a non-empty set  $X$ . By order separability, there exists a set  $Z \subseteq X$  such that for any  $a, b \in X$  with  $a \succ b$ , there exists  $z \in Z$  satisfying  $a \succsim z \succsim b$ .
- From order separability, we start by constructing indifference classes based on adjacent elements. First, we can define a set  $P(z)$  of immediate strict predecessors for each  $z \in Z$ , such that each element of  $P(z)$  is strictly preferred to  $z$  but with no intermediate elements between them, ie  

$$P(z) = \{a \in X \mid a \succ z \text{ and } \nexists w \in X \text{ such that } a \succ w \succ z\}$$
  - If  $P(z)$  is non-empty, all elements within  $P(z)$  must be indifferent to each other. Therefore they form an indifference class within  $X$ , such that we can select one representative element from  $P(z)$  and call it  $u_z$ . Let  $U$  be the set of all such selected representatives. Because there is at most one  $u_z$  for each  $z \in Z$ , and  $Z$  is countable,  $U$  is strictly countable.
- Symmetrically, define the set of immediate strict successors,  

$$S(z) = \{b \in X \mid z \succ b \text{ and } \nexists w \in X \text{ such that } z \succ w \succ b\}$$
such that all elements in  $S(z)$  are indifferent to each other. Select exactly one representative element from  $S(z)$  and call it  $v_z$ . Let  $V$  be the set of all such selected representatives and by the same logic as above,  $V$  is countable.

- Define  $Y = Z \cup U \cup V$ . Because  $Y$  is the union of three countable sets,  $Y$  is countable.
- Now that we have separated  $X$  into indifference classes, we can proceed with the main proposition. We have shown that there exists a countable set  $Y \subset X$ , defined as above. Let  $x, z \in X$  such that  $x \succ z$ . We must prove there exist  $y_1, y_2 \in Y$  such that  $x \succsim y_1 \succ y_2 \succsim z$ .
- By order separability of  $x \succ z$ , there exists some  $z_1 \in Z$  such that  $x \succsim z_1 \succsim z$ . Because  $x \succ z$ , we cannot have both  $x \sim z_1$  and  $z_1 \sim z$ . Therefore, at least one of  $x \succ z_1$  or  $z_1 \succ z$  must hold. By cases:
  - If  $x \succ z_1$ , we can apply order separability again to  $x \succ z_1$  to show that there exists  $z_2 \in Z$  such that  $x \succsim z_2 \succ z_1$ .
    - If  $z_2 \succ z_1$ , then we can define  $y_1 = z_2$  and  $y_2 = z_1$  and we have  $x \succsim z_2 \succ z_1 \succsim z$ , so the proof is satisfied.
    - If instead  $\forall z' \in Z$  such that  $x \succsim z' \succ z_1$ , we have  $z' \sim z_1$ , we can show that there is no element  $w \in X$  such that  $x \succ w \succ z_1$ . If there was such a  $w$ , order separability on  $x \succ w$  would guarantee a  $z_w \in Z$  where  $x \succsim z_w \succ w$ . Since  $w \succ z_1$ , that would imply  $z_w \succ z_1$ . This contradicts the premise that all such elements in  $Z$  between  $x$  and  $z_1$  are indifferent to  $z_1$ , therefore no such  $w$  can exist. Thus  $x$  is an immediate predecessor to  $z_1$ , meaning  $x \in P(z_1)$ , and  $U$  must have  $u_{z_1} \in U \subseteq Y$ . Because  $x, u_{z_1} \in P(z_1)$ , they are indifferent, so  $x \sim u_{z_1}$ . Let  $y_1 = u_{z_1}$  and  $y_2 = z_1$  such that we can show  $x \sim u_{z_1} \succ z_1 \succsim z$ , satisfying the proof.
  - If  $z_1 \succ z$ , we can apply order separability to  $z_1 \succ z$  to show that there exists  $z_3 \in Z$  such that  $z_1 \succsim z_3 \succ z$ .
    - If  $z_1 \succ z_3$ , then immediately we have  $y_1 = z_1$  and  $y_2 = z_3$  such that  $x \succsim z_1$ , and we have  $x \succsim z_1 \succ z_3 \succ z$ , satisfying the proof.
    - If instead we have  $z' \sim z_1$  for every  $z' \in Z$  satisfying  $z_1 \succsim z' \succ z$ , then we can apply the same logic as above to

show that there can be no  $w \in X$  such that  $z_1 \succ w \succ z$ . If  $w$  existed, order separability applied to  $w \succ z$  would imply  $z_w \in Z$  such that  $w \succsim z_w \succsim z$ . Because  $z_1 \succ w$ , this forces  $z_1 \succ z_w$ , contradicting that all elements in  $Z$  between  $z_1$  and  $z$  are indifferent to  $z_1$ . Because no such  $w$  exists,  $z$  is an immediate successor to  $z_1$ , meaning  $z \in S(z_1)$  and our definition of  $V$  guarantees  $v_{z_1} \in V \subseteq Y$ . Because  $z, v_{z_1} \in S(z_1)$ , they are indifferent, so  $v_{z_1} \sim z$ . Finally, let  $y_1 = z_1$  and  $y_2 = v_{z_1}$ , so that we have  $x \succsim z_1 \succ v_{z_1} \sim z$ , satisfying the proof.

- In all cases, for any  $x \succ z$ , the intermediate elements  $y_1, y_2 \in Y$  are guaranteed to exist within  $Y$  such that  $x \succsim y_1 \succ y_2 \succsim z$ , completing the proof.

Let  $X$  be a non-empty set and  $\succsim$  be an order separable preference relation.

2. Let  $Y$  be such a set numbered  $\{y_1, y_2, \dots\}$ . Define

$L(x) = \{m \in \mathbb{N} \mid x \succ y_m\}$ . Define  $u : X \rightarrow [0, 1]$  as

$$u(x) = \begin{cases} 0 & \text{if } L(x) \text{ is empty} \\ \sum_{m \in L(x)} \frac{1}{2^m} & \text{otherwise} \end{cases}$$

Show  $u$  represents  $\succsim$ .

- To prove that  $u : X \rightarrow [0, 1]$  represents the preference relation  $\succsim$ , we must establish that for any  $x, z \in X$ ,  $x \succsim z \iff u(x) \geq u(z)$ . Because  $\succsim$  is a complete and transitive relation, this is equivalent to proving that  $x \succsim z \implies u(x) \geq u(z)$  and that  $x \succ z \implies u(x) > u(z)$
- First we show that weak preference implies subset inclusion. Assume  $x \succsim z$ . Let  $m \in L(z)$ . By the definition of the set  $L$ , this implies  $z \succ y_m$ . Because  $\succsim$  is transitive,  $x \succsim z$  and  $z \succ y_m$  implies  $x \succ y_m$ . Therefore,  $m \in L(x)$ .
- Since any element in  $L(z)$  must also be in  $L(x)$ , we know  $L(z)$  is a subset of  $L(x)$ ,  $L(z) \subseteq L(x)$
- Since  $L(z)$  is a subset of  $L(x)$ , the summation over  $L(x)$  must be greater than or equal to the summation over  $L(z)$ :

$$\sum_{m \in L(x)} \frac{1}{2^m} \geq \sum_{m \in L(z)} \frac{1}{2^m} \implies u(x) \geq u(z)$$

- If  $x \sim z$ , symmetric logic yields  $L(x) \subseteq L(z)$  and  $L(z) \subseteq L(x)$ , meaning  $L(x) = L(z)$  and  $u(x) = u(z)$ , so indifference mapping holds.
- To show that strict preferences also imply subset inclusion, we first assume  $x \succ z$ . We know there exist  $y_1, y_2 \in Y$  such that  $x \succ y_1 \succ y_2 \succ z$ . Let  $k \in \mathbb{N}$  be the index corresponding to the element  $y_2$  in  $Y = \{y_1, y_2, \dots\}$ , such that  $y_2 = y_k$ .

- By transitivity,  $x \succsim y_1 \succ y_k$  implies  $x \succ y_k$ . Therefore,  $k \in L(x)$ .  
However, we also know  $y_k \succsim z$ , therefore we cannot have that  $z \succ y_k$ .  
Therefore,  $k \notin L(z)$ . Thus we have that  $L(z) \subseteq L(x)$ , but since there is at least one  $k$  such that  $k \in L(x)$  and  $k \notin L(z)$ , we know that  $L(z)$  is a strict subset of  $L(x)$   $L(z) \subset L(x)$
- Because the term  $\frac{1}{2^k}$  is strictly positive, the strict subset inclusion forces a strictly greater summation:

$$\sum_{m \in L(x)} \frac{1}{2^m} = \sum_{m \in L(z)} \frac{1}{2^m} + \frac{1}{2^k} + \sum_{m \in L(x) \setminus (L(z) \cup \{k\})} \frac{1}{2^m}$$

$$u(x) \geq u(z) + \frac{1}{2^k} > u(z)$$

- Therefore since  $x \succsim z \implies u(x) \geq u(z)$  and  $x \succ z \implies u(x) > u(z)$ , we are guaranteed that  $u(x) \geq u(z) \iff x \succsim z$  and thus  $u(x)$  rationalizes the preference relation.

## Question 4:

Consider a data set that allows us to observe not only what a DM 'finally' chooses, but also how their choice changes with contemplation time.

Specifically, define the following:

- $X$ : Finite grand choice set
- $\mathcal{X}$ : Non-empty subsets of  $X$
- $Z \in \{Z_t\}_{t=1}^{\infty}$ : Sequences of elements of  $X$
- $\mathcal{Z}$ : set of sequences  $Z$
- $\mathcal{Z}_A \subset \mathcal{Z}$ : set of sequences s.t.  $Z_t \subset A \in \mathcal{X}$

Define a data set of the following type:

**Definition 1** A *Choice Process Data Set*  $(X, C)$  comprises of:

- finite set  $X$
- choice function  $C : \mathcal{X} \rightarrow \mathcal{Z}$

such that  $C(A) \in \mathcal{Z}_A \quad \forall A \in \mathcal{X}$ .

**Definition 2** A *model alternative based search* consists of a utility function  $u : X \rightarrow \mathbb{R}$  and a search correspondence

$$S : \mathcal{X} \rightarrow \mathcal{Z}$$

with  $S(A) \in \mathcal{Z}_A \quad \forall A \in \mathcal{X}$

Such that the DM always chooses best option of those searched:

$$C_A(t) = \arg \max_{x \in S_A(t)} u(x)$$

1. Show that the model as stated places no restriction on the data set.
2. Add the assumption that  $S_A(t) \subseteq S_A(t + s)$ . Show that this must mean, that for every set  $A$ , there exists a  $t$  such that, for  $s > t$ ,

$C(A, t) = C(A, s)$ . Define this set as  $\bar{C}(A)$  for each  $A$ .

3. Does the above model place any restrictions on the behavior of  $\bar{C}$ ? If not, can you propose any additional restrictions on  $S$  that would ensure that  $\bar{C}$  must obey properties  $\alpha$  and  $\beta$ ?
4. Go back to the original model described in part (2) above. What behavior in this model would imply that  $u(x) > u(y)$ ? What behavior would imply  $u(x) = u(y)$ ?
5. Write down a condition that guarantees that we can find a utility function that represents the 'revealed preference' information from part (4).
6. **Hard** Show that this condition is necessary and sufficient for a data set to be consistent with the above model.

## Question 4 Solutions:

Consider a data set that allows us to observe not only what a DM 'finally' chooses, but also how their choice changes with contemplation time.

Specifically, define the following:

- $X$ : Finite grand choice set
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Such that the DM always chooses best option of those searched:

$$C_A(t) = \arg \max_{x \in S_A(t)} u(x)$$

1. Show that the model as stated places no restriction on the data set.

- We aim to show that any arbitrary data set  $(X, C)$  can be rationalized by at least one utility function  $u$  and search correspondence  $S$ . Thus,

let  $(X, C)$  be an arbitrary data set where  $C(A) \in \mathcal{Z}_A$  for all  $A \in \mathcal{X}$ .

- Since we only need to demonstrate that the model works for at least one utility function, not all functions, we can define the utility function  $u : X \rightarrow \mathbb{R}$  such that  $u(x) = 0$  for all  $x \in X$ .
- We can define the search correspondence  $S : \mathcal{X} \rightarrow \mathcal{Z}$  such that for every choice set  $A \in \mathcal{X}$  and at every time period  $t$ , the searched set is the same as the choice  $S_A(t) = C_A(t)$
- We are guaranteed  $C(A) \in \mathcal{Z}_A$ , so our correspondence satisfies  $S(A) \in \mathcal{Z}_A$  for all  $A \in \mathcal{X}$ .
- Given  $C_A(t) = \arg \max_{x \in S_A(t)} u(x)$ , we can substitute  $S_A(t)$  with  $C_A(t)$  such that  $\arg \max_{x \in C_A(t)} u(x)$ . Because the utility function  $u(x)$  is constant across all  $X$ , every element equally maximizes  $u$ . Thus the subset of  $C_A(t)$  that maximizes utility is all of  $C_A(t)$ .
- We can show that this satisfies  $C_A(t) = \arg \max_{x \in S_A(t)} u(x)$  for any possible sequence of choices. Thus the model places no restrictions on the dataset.

2. Add the assumption that  $S_A(t) \subseteq S_A(t + s)$ . Show that this must mean, that for every set  $A$ , there exists a  $t$  such that, for  $s > t$ ,  $C(A, t) = C(A, s)$ . Define this set as  $\bar{C}(A)$  for each  $A$ .
- Since  $S_A(t) \subseteq S_A(t + s)$  we have that each subsequent choice set is at least non-decreasing, since the supersets  $S_A(t + s)$  also search over the previously considered elements, such that  $\{S_A(t)\}_{t=1}^{\infty}$  is monotonically increasing. Since the set  $X$  is finite, any subset  $A \in \mathcal{X}$  must also be finite, such that the total search space will eventually be exhausted by a sufficiently large sequence of searches. We can define the time  $t^*$  the point in the sequence where the search has explored the entire finite set of options, such that for all  $s > t^*$ ,  $S_A(s) = S_A(t^*)$
  - The model specifies that  $C_A(t) = \arg \max_{x \in S_A(t)} u(x)$ , but since the search set stabilizes to  $S_A(t^*)$  after time  $t^*$ , we know that  $C(A, s) = \arg \max_{x \in S_A(s)} u(x) = \arg \max_{x \in S_A(t^*)} u(x) = C(A, t^*)$  for all  $s > t^*$ , which we can define as  $\bar{C}(A)$ .
3. Does the above model place any restrictions on the behavior of  $\bar{C}$ ? If not, can you propose any additional restrictions on  $S$  that would ensure that  $\bar{C}$  must obey properties  $\alpha$  and  $\beta$ ?
- The model does not impose any restrictions on the behavior of correspondence  $\bar{C}$  across different choice sets, which allows violations of Axioms  $\alpha$  and  $\beta$ .
  - Because the mapping from a choice set  $A$  to its search set  $\bar{S}(A) = \bigcup_{t=1}^{\infty} S_A(t)$  is arbitrary, the terminal sets may not satisfy the axioms. We can show this via the following cases; first assume a utility function  $u(x) > u(y) > u(z)$  and sets  $A = \{x, y, z\}$  and  $B = \{x, y\}$ , such that  $B \subseteq A$ .
    - This case violates  $\alpha$ , since the search process can arrive at sets like  $\bar{S}(A) = \{x, z\}$  and  $\bar{S}(B) = \{y\}$ . Maximizing utility on these

sets results in  $\bar{C}(A) = \{x\}$  and  $\bar{C}(B) = \{y\}$ , which means that we have  $x \in B \subseteq A$  and  $x \in \bar{C}(A)$ , but  $x \notin \bar{C}(B)$  because the search did not include  $x$  in subset  $B$ .

- This case also violates  $\beta$ . If we change the (arbitrary) utility function such that  $u(x) = u(y) > u(z)$ , we can construct a search that results in  $\bar{S}(B) = \{x, y\}$  and  $\bar{S}(A) = \{y, z\}$ . The correspondence results in  $\bar{C}(B) = \{x, y\}$  and  $\bar{C}(A) = \{y\}$ , which violates axiom  $\beta$  since  $x, y \in \bar{C}(B)$ ,  $B \subseteq A$ , and  $y \in \bar{C}(A)$ , but  $x \notin \bar{C}(A)$ .
- To ensure that  $\bar{C}$  satisfies  $\alpha$  and  $\beta$ , we can impose the following restriction. For any  $B \subseteq A$ , if we require  $\bar{S}(B) = \bar{S}(A) \cap B$ , then any element considered in the superset is considered in its subsets. The intersection property, which is similar to WARP, induces the axioms as follows:
  - To show  $\alpha$ : Let  $x \in B \subseteq A$  and  $x \in \bar{C}(A)$ . By definition,  $x \in \bar{S}(A)$  and  $x$  maximizes  $u$  on  $\bar{S}(A)$ . Because  $x \in B$ , the restriction requires  $x \in \bar{S}(A) \cap B = \bar{S}(B)$ . Because  $x$  maximizes  $u$  for superset  $\bar{S}(A)$ , it must maximize  $u$  in any subset containing  $x$ , including  $\bar{S}(B)$ . Therefore,  $x \in \bar{C}(B)$ .
  - To show  $\beta$ : Let  $x, y \in \bar{C}(B)$ ,  $B \subseteq A$ , and  $y \in \bar{C}(A)$ . Because both  $x$  and  $y$  are chosen in  $B$ ,  $u(x) = u(y)$ . Because  $y \in \bar{C}(A)$ ,  $y \in \bar{S}(A)$  and  $y$  maximizes  $u$  on  $\bar{S}(A)$ . Because  $x \in \bar{C}(B)$ , we know  $x \in \bar{S}(B)$ . The restriction requires that since  $x \in \bar{S}(B)$ , it must be that  $x \in \bar{S}(A)$ . Since  $u(x) = u(y)$ ,  $x$  is equally preferred to  $y$  in  $\bar{S}(A)$ , which means  $x \in \bar{C}(A)$ .
- Thus both axioms are satisfied given the intersection restriction.

4. Go back to the original model described in part (2) above. What behavior in this model would imply that  $u(x) > u(y)$ ? What behavior would imply  $u(x) = u(y)$ ?
- Given the expanding search sets  $S_A(t) \subseteq S_A(t + s)$  model from part (2), we can show that the sequence of choices reveals utility by showing which elements are kept and which are discarded in neighboring nested sets.
  - To show when  $u(x) > u(y)$ , we simply check when a new element  $x$  is chosen over the previous utility maximizing choice  $y$ . Given an initial choice of  $y$  from a set  $A \in \mathcal{X}$  at time  $t_1$ , we know that  $y \in C_A(t_1)$  such that  $y \in S_A(t_1)$ . By  $S_A(t_1) \subseteq S_A(t_2)$  we know that  $y$  remains in the search set at a later time  $t_2$ . Thus, since the chosen elements  $C_A(t_2)$  maximize the utility function  $u$  over  $S_A(t_2)$ , if we observe  $x \in C_A(t_2)$  but  $y \notin C_A(t_2)$ , with  $x \neq y$ , then we know that  $u(x) > u(y)$ .
  - To show when  $u(x) = u(y)$ , we need to observe a simultaneous multi-valued choice. If we observe both  $x$  and  $y$  are chosen concurrently from the same set  $A \in \mathcal{X}$  at the same time period  $t$  such that  $x, y \in C_A(t)$ , then we know that  $x, y \in S_A(t)$  and they must both have an identical supremum of the utility function. Therefore,  $u(x) = u(y)$ .
5. Write down a condition that guarantees that we can find a utility function that represents the 'revealed preference' information from part (4).
- To ensure the utility function represents the preference information, we must have that  $x \succsim y \implies u(x) \geq u(y)$  and  $x \succ y \implies u(x) > u(y)$ .
  - If we take  $S_A(t_1) \subseteq S_A(t_2)$  for any  $t_1 \leq t_2$  from part (2) then we know that any option searched at  $t_1$  remains available at  $t_2$ .
  - If we take the weak preference relation  $\succsim$  on the finite set  $X$  as  $x \succsim y \iff \exists A \in \mathcal{X} \text{ and } t_1 \leq t_2 \text{ such that } y \in C_A(t_1) \text{ and } x \in C_A(t_2)$ ,

then if  $y$  was in the choice set at  $t_1$ , it remains in the search set at  $t_2$ . If  $x$  is chosen at  $t_2$  when  $y$  is available then we know that  $x$  is at least as good as  $y$ .

- If we then define the strict preference relation  $\succ$  on  $X$  as  $x \succ y \iff \exists A \in \mathcal{X} \text{ and } t_1 < t_2 \text{ such that } y \in C_A(t_1), x \in C_A(t_2), \text{ and } y \notin C_A(t_2)$ , then if  $y$  is dropped from the choice correspondence and replaced with  $x$  as  $t$  increases, we know there is a strict preference for  $x$ .
- The condition that guarantees a utility representation for this model and preference relation is the Generalized Axiom of Revealed Preferences (GARP). It states that transitive closure of weak preference relations cannot contain any contradictory cycles, ie for any finite sequence  $x_1, x_2, \dots, x_k \in X$  where  $x_1 \succsim x_2, x_2 \succsim x_3, \dots, x_{k-1} \succsim x_k$ , we cannot have  $x_k \succ x_1$ , thus it must hold that  $x_1 \succsim x_k$ .
- For finite  $X$ , satisfying GARP guarantees acyclic preferences and thus we have a complete, transitive weak order that extends the preference relation. Thus, there must exist a real-valued utility function  $u : X \rightarrow \mathbb{R}$  such that  $x \succsim y \implies u(x) \geq u(y)$  and  $x \succ y \implies u(x) > u(y)$ .
- If, on the other hand, all elements are equally indifferent, such that  $x \sim y \forall x, y \in X$  then we have trivially that  $x \succsim y \implies u(x) = u(y)$  which is still rationalizable for a constant utility function (ie  $u(x) = 0 \forall x \in X$ ).
- Therefore, either way, we have a condition that guarantees a utility function that represents the revealed preferences.

6. **(Hard)** Show that this condition is necessary and sufficient for a data set to be consistent with the above model.

- To prove that GARP is necessary and sufficient for a data set to be consistent with the above model, and thereby there is strict acyclicity in the preference relations, we consider both sides of the proof:
- Necessity (Consistent Model  $\implies$  GARP): Assume the data set is consistent with the model.
  - Therefore, there exists a utility function  $u : X \rightarrow \mathbb{R}$  and a search correspondence  $S_A(t) \subseteq S_A(t + s)$  such that  $C_A(t) = \arg \max_{x \in S_A(t)} u(x)$  for all  $A \in \mathcal{X}$ .
  - Assume for contradiction that the data contains a strict cycle  $x_1 \succsim x_2, x_2 \succsim x_3, \dots, x_{k-1} \succsim x_k$ , and  $x_k \succ x_1$ . Then we would have the following inequalities  $u(x_1) \geq u(x_2) \geq u(x_3) \cdots \geq u(x_k) > u(x_1)$ , but that implies that  $u(x_1) > u(x_1)$ , which is impossible. Thus we have arrived at a contradiction, so the data must not contain any strict cycles that violate GARP. Thus GARP is satisfied.
- Sufficiency (GARP  $\implies$  Consistent Model): Assume the data set satisfies GARP. We must construct a utility function  $u$  and search correspondence  $S$  that generates the observed choices of the model.
  - Because  $X$  is finite and the preference relations  $\succsim$  and  $\succ$  satisfy strict acyclicity, we know they form a complete, transitive weak order on  $X$  that extends the preference relation. Thus, there must exist a real-valued utility function  $u : X \rightarrow \mathbb{R}$  such that  $x \succsim y \implies u(x) \geq u(y)$  and  $x \succ y \implies u(x) > u(y)$
  - For any set  $A \in \mathcal{X}$  and time  $t$ , define the search set as the union of all elements chosen from  $A$  up to and including time  $t$  such that we have  $S_A(t) = \bigcup_{s=1}^t C_A(s)$ . Because  $C_A(s) \subseteq A$  for all  $s$ ,  $S_A(t) \subseteq A$ , satisfying the restriction  $S(A) \in \mathcal{Z}_A$ . Since we have an

ongoing union, we have that  $S_A(t) \subseteq S_A(t+1)$ , satisfying our condition from part (2).

- Thus we have to prove that  $C_A(t) = \arg \max_{x \in S_A(t)} u(x)$ , which is the same as proving that  $C_A(T) = S_A(t)$ :
  - If all chosen elements are maximizers, such that for  $x \in C_A(t)$  and alternative  $y \in S_A(t)$ , we know that  $S_A(t)$ ,  $y \in C_A(s)$  for some time  $s \leq t$ . Since we observe  $y \in C_A(s)$  and  $x \in C_A(t)$  with  $s \leq t$ , we know that  $x \succsim y$  and thus  $u(x) \geq u(y)$ . Therefore  $x$  maximizes  $u$  over the set  $S_A(t)$  and  $C_A(t) \subseteq S_A(t)$ .
  - If all maximizers are chosen elements, such that if  $z \in S_A(t)$  maximizes  $u$  over  $S_A(t)$ , then we know that  $u(z) = u(x)$  for the chosen element  $x \in C_A(t)$ . Because  $z \in S_A(t)$ ,  $z \in C_A(s)$  for some  $s \leq t$ .
    - If we assume for contradiction that  $z \notin C_A(t)$ , then that implies  $s < t$ . The observation of  $z \in C_A(s)$ ,  $x \in C_A(t)$ , and  $z \notin C_A(t)$  satisfies the GARP condition that  $x \succ z$ . However,  $x \succ z$  requires  $u(x) > u(z)$ , which contradicts our assumption that  $z$  is a maximizer ( $u(z) = u(x)$ ).
    - Thus our assumption  $z \notin C_A(t)$  must be false such that any element maximizing  $u$  on  $S_A(t)$  is necessarily contained in  $C_A(t)$ , ie that  $S_A(t) \subseteq C_A(t)$ .
- Since we have shown  $S_A(t) \subseteq C_A(t)$  and  $C_A(t) \subseteq S_A(t)$ , we know that  $C_A(t)$  is equal to the set of utility-maximizing elements in  $S_A(t)$ , satisfying the model conditions.